

TermPlanMT: Graph-Grounded Terminology Planning for French–English Regulatory Medical Document Translation

When Ontology Grounding Helps and When It Does Not

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Abstract. Terminology consistency is a critical quality dimension in the machine translation of regulatory medical documents, where mistranslated adverse event terms carry direct patient-safety consequences. We present **TermPlanMT**, a graph-augmented terminology-planning framework evaluated on a 126-segment French–English corpus drawn from a European Medicines Agency (EMA) Summary of Product Characteristics (SmPC). Six systems form an ablation ladder isolating the contributions of NER recall, MedDRA ontology grounding, and constrained decoding. Our central finding is that Concept Coverage Rate (CCR), the fraction of source spans grounded in a MedDRA ontology graph, is the binding constraint on terminology-aware MT: at 39% CCR, graph-free systems (S1–S2) outperform all graph-augmented systems on fluency – S1 leads on BLEU, S2 on chrF and Hypothesis–Reference Agreement (HRA) – while improving NER quality yields ≈ 7 BLEU gains for graph-augmented Mistral systems (S3–S5-Mistral) but not for NLLB or S2. HTM scores the certified human reference at 0.15–0.18, unchanged by full MedDRA alias expansion, confirming a structural mismatch between MedDRA nomenclature and professional translation practice. Ambiguous grounding causes the planning stage to abstain from locking critical terms, explaining why graph-augmented systems produce lower cross-document consistency than graph-free baselines.

1 Introduction

EMA Summaries of Product Characteristics (SmPCs) require a single adverse-event concept to receive one identical English rendering across an entire document—a requirement that affects pharmacovigilance signal detection and patient safety [Neves et al., 2022]. Neural MT handles medical prose fluently but is structurally indifferent to cross-segment consistency, producing term drift (same source term, multiple renderings), concept flattening (Lowest Level Term (LLT)-level terms collapsed to hypernyms), and coverage gaps (multi-token spans outside NER). MedDRA’s five-level hierarchy (Table 1) provides validated English labels that are an attractive grounding resource—if the grounding pipeline can cover enough of the source text.

This paper makes the following contributions:

1. **A six-system ablation ladder** that isolates the individual contributions of NER recall, MedDRA ontology grounding via Neo4j, and constrained decoding enforcement to terminology-aware MT quality.
2. **Three novel evaluation metrics** – HTM, rHTM, and HRA – that measure terminology alignment at the ontology level rather than surface string overlap, with a diagnostic showing $rHTM \approx 0.15\text{--}0.18$ for certified human references.
3. **A CCR bottleneck analysis** demonstrating that NER extraction quality, not retrieval or decoding strategy, is the primary constraint on graph-augmented MT performance.
4. **A mechanistic consistency analysis** explaining the paradox in which graph-augmented systems produce lower cross-document term consistency than graph-free baselines, traced to the planning stage’s abstain-on-ambiguity policy.
5. **Reproducible artefacts** including per-segment HTM scores, bootstrap confidence intervals, and qualitative drift analysis exported as CSV and Markdown.
6. **Two named phenomena:** the **register mismatch** (graph-augmented systems produce pharmacovigilance coding language while certi-

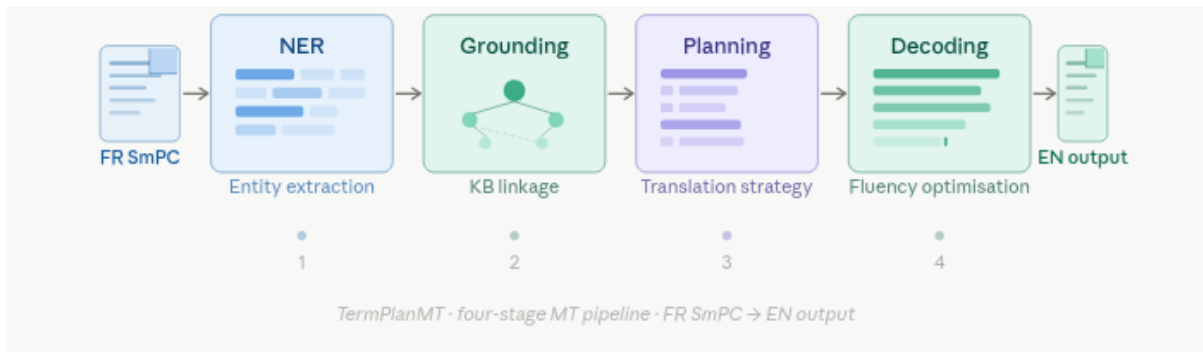


Figure 1: Four-stage TermPlanMT pipeline (NER → grounding → planning → decoding).

fied translators write clinical prose) and the **consistency paradox** (the most ontology-correct system is the least human-like and is never the preferred output in 30-segment human evaluation).

2 Related Work

Lexically constrained decoding [Hokamp and Liu, 2017, Post and Vilar, 2018] and terminology-aware training [Dinu et al., 2019] enforce surface-level term appearance but ignore ontological hierarchy and do not address candidate selection when multiple levels are valid. Herold and Ney [2023] show that feeding long context directly into NMT degrades output quality due to attention mechanism limitations, motivating our pre-translation planning stage: rather than exposing the full document at decoding time, we distil relevant terminology into a compact, structured prompt. Labrak et al. [2024] release BioMistral, the medically adapted LLM used for NER here. Neves et al. [2022] identify biomedical named entities broadly—diseases, drugs, and procedures—as a persistent failure mode in biomedical MT. Our work narrows this to the specific subtype of adverse event (AE) terminology as defined by the MedDRA ontology, a distinction with direct regulatory consequences given that AE terminology is legally standardised across EMA submissions. GraphRAG-style retrieval [Edge et al., 2024] into LLM prompts provides the template for our S3 system, adapting knowledge-grounded MT [Moussallem et al., 2019] to a Neo4j MedDRA backend.

3 Dataset

Corpus.. Section 4.8 (Undesirable Effects) of the KEYTRUDA (pembrolizumab) SmPC [European Medicines Agency, 2024]: 126 aligned French–English segment pairs after excluding segment 48_028 (a dense frequency table whose n-

gram overlap distorts corpus metrics). Coverage: $\approx 4,400$ words; all three failure modes at high density.

Level	Abbr.	Example
L1	SOC	Respiratory disorders
L2	HLGT	Pulmonary conditions
L3	HLT	Pneumonitis
L4	PT	Pembro. pneumonitis
L5	LLT	Immune-med. pneumonitis

Table 1: MedDRA hierarchy. Adverse-event mentions in clinical text map almost exclusively to L5 ($\approx 96\%$).

NER conditions as a CCR perturbation..

Rather than comparing two NER systems, we use them as a controlled perturbation of CCR within the same pipeline. **Baseline NER** (ner_biollm): BioMistral-7B zero-shot; CCR = 39.0%, 184 grounded spans across 472 extracted (126-segment evaluation set)¹. This condition exhibits a systematic artefact—the model generates *hypothyroidie* for 92/127 segments regardless of content—partially contaminating consistency measurements. **Fine-tuned (higher-quality) NER** (ner_biollm_ft): BioMistral-7B fine-tuned on a hierarchical MedDRA SFT dataset; 211 grounded spans across 580 extracted (36.4% CCR) with cleaner boundaries and no hallucinated terms. Graph-augmented systems benefit substantially under this condition, confirming that *effective* span quality (clean boundaries, usable grounding)—not raw graph retrieval—is

¹ CCR is a joint function of NER recall and graph coverage. The fine-tuned NER extracted more spans (580 vs. 472) yet produced lower CCR (36.4% vs. 39.0%) because stronger NER finds rarer, harder terms less likely to be in the graph. The binding constraint is therefore absolute grounded span count (211 vs. 184), not CCR alone.

the binding constraint once contamination is controlled.

4 System Architecture

Six systems form an ablation ladder (Table 2). S1–S5 share the same four-stage pipeline: French NER → MedDRA grounding via Neo4j → global terminology lock → MT decoding. S6 is a glossary oracle ablation that replaces the MedDRA graph with a curated FR→EN glossary extracted from the reference translations.

ID	Model	Graph	Enforce.
S1	NLLB-200 600M	No	None
S2	Mistral-7B (doc)	No	None
S3	Mistral+GraphRAG	Yes	None
S4	S3 + reranking	Yes	Soft
S5	NLLB/Mistral+logit	Yes	Hard
S6	NLLB + glossary	Oracle	Hard

Table 2: Ablation ladder. S1 = NMT lower bound; S2 = LLM with full document context; S3 is the key novel condition; S6 isolates term-source quality from graph retrieval.

Terminology planning.. Before translation, `compute_global_locks()` scores each candidate English rendering c :

$$\text{Score}(c) = \alpha \text{Ctx}(c) + \beta \text{Nbr}(c) - \gamma \text{HP}(c) \quad (1)$$

where Ctx is cosine similarity to source segments containing the term, Nbr is similarity to adjacent MedDRA concept renderings, and $\text{HP} = |\hat{d} - d_{\text{src}}|$ penalises wrong specificity level ($\alpha=0.5$, $\beta=0.3$, $\gamma=0.2$). When a term matches *multiple* MedDRA concepts, no lock is committed (§8).

The abstain-on-ambiguity policy (line 4) is a deliberate safety choice: when a French term maps to multiple MedDRA concepts with equal confidence (e.g. *pneumopathie inflammatoire* maps to both PNEUMONITIS and INTERSTITIAL LUNG DISEASE), forcing a lock risks propagating the wrong concept across the entire document. The cost of this conservatism is reduced cross-document consistency, since unlocked terms are resolved independently at each segment by the decoder. We discuss this consistency paradox in Section 8.

S5 – Logit Boosting with Hard Fallback.. S5 applies a two-layer decoding strategy. First, a

Algorithm 1 Global Terminology Planning – `compute_global_locks`

```

1 # Inputs: segments S, MedDRA graph G
2 # Weights: alpha=0.5, beta=0.3, gamma=0.2
3 # Output: lock map {fr_term -> en_rendering}
4
5 locks = {}
6 for fr_term in unique_ner_terms(S):
7     concept = G.ground(fr_term)
8     # exact match first, then fuzzy >= 90%
9
10    if concept is None or G.is_ambiguous(
11        fr_term):
12        continue # abstain: wrong lock
13                propagates document-wide
14
15    candidates = [concept.name] + G.
16                neighbours(concept)
17    best, best_score = None, -inf
18    for r in candidates:
19        ctx = cosine(embed(segs), embed(r))
20        nbr = cosine(embed(r), embed(G.
21                neighbours(concept)))
22        hp = abs(level(r) - level(concept))
23        score = alpha*ctx + beta*nbr - gamma*
24                hp
25        if score > best_score:
26            best, best_score = r, score
27    locks[fr_term] = best
28
29 return locks

```

soft logit bias (PhraseLogitBoost, $\delta=1.75$ for NLLB, $\delta=1.25$ for Mistral) increases the probability mass of tokens belonging to locked MedDRA renderings without preventing the model from generating alternative phrasings if contextually motivated. Second, for NLLB only, `force_words_ids` enforces the locked phrase as a hard constraint via constrained beam search, with automatic fallback to soft-only decoding if the constraint causes beam failure. The Mistral variant uses soft boosting only, since instruction-tuned LLMs are more sensitive to hard token forcing and exhibit greater fluency degradation under strict constraints.

4.1 MedDRA Knowledge Graph Construction

The grounding component $\mathcal{G}$ used throughout Algorithms 1–5 is a Neo4j property graph built from the MedDRA v28 flat-file release, obtained under academic licence via UPV/EHU. Construction proceeds in four stages.

(1) Parsing the ASCII flat files.. MedDRA distributes five pipe-delimited *.asc files. The primary source is `mdhier.asc`, where each row encodes the full SOC → HLGT → HLT → PT → LLT

lineage for one concept, together with English and French preferred labels. A custom parser handles both the pipe-separated format (v28+) and the legacy dollar-delimited format, with automatic encoding detection (UTF-8 → CP1252 → ISO-8859-1). This yields approximately 80,000 LLT records with complete five-level lineage.

(2) Building the French label index.. The French MedDRA release is read in parallel to populate a `fr_label` property on each concept node. After case-folding normalisation, 7,679 French surface forms are *ambiguous* — a single normalised string maps to two or more distinct MedDRA concepts (e.g., *pneumopathie inflammatoire* resolves to both PNEUMONITIS and INTERSTITIAL LUNG DISEASE). These keys are recorded in an `_ambiguous` set and trigger the abstain-on-ambiguity policy in `compute_global_locks` (Algorithm 1).

(3) Loading into Neo4j.. Each MedDRA concept is stored as a `:Concept` node with properties `{id, name, fr_label, level, tier}`. Parent–child relationships are represented as directed `BROADER_THAN` edges, producing a five-level DAG. The graph is populated via batched Cypher `MERGE` transactions against a Neo4j 5 Community instance deployed via Docker Compose. The final graph contains **84,404 concept nodes** and **91,082 entries** in the PT/LLT fuzzy index.

(4) Runtime API (TermGraph).. All French concept keys are loaded into memory on first call (lazy, cached). String grounding applies exact case-fold matching first; if that fails, RapidFuzz cosine similarity with a threshold of $\geq 90\%$ is applied, restricted to PT- and LLT-level nodes only (higher levels produce over-generalised matches). The `neighbours()` method returns concepts reachable via one or two `BROADER_THAN` hops; `same_branch()` executes a `shortestPath` Cypher query bounded to five hops. A CCR diagnostic over the baseline NER condition confirms the grounding ceiling: 23% EXACT-HIT, 11% FUZZY-HIT, 6% AMBIGUOUS, and 60% NOT-IN-GRAPH — the last category dominated by drug names, grade annotations, and numeric tokens that lie outside the adverse-event subset of MedDRA indexed in the graph.

5 Metrics

Metric definitions.

- **HTM** (Hierarchy-Aware Terminology Metric): per grounded NER span, score = 1.0 (exact MedDRA level), 0.5 (same branch), 0.0 (miss); macro-averaged (Algorithm 2).
- **rHTM** (Reference HTM): HTM applied to the certified human reference, not a system hypothesis. A dataset-level ceiling; values of 0.15–0.18 confirm translators prefer phrasings outside the MedDRA preferred-term vocabulary.
- **HRA** (Hypothesis–Reference Agreement): $1 - |\text{HTM}_{\text{hyp}} - \text{HTM}_{\text{ref}}|$, macro-averaged (Algorithm 3). Measures agreement with professional translation practice rather than raw ontology proximity.
- **HTM-A** (Alias-Expanded HTM): accepts any ancestor up to SOC. Score invariance across tiers T1–T3 confirms failures are conceptual, not surface-level.
- **CCR** (Concept Coverage Rate): grounded spans / extracted spans. Identical across S1–S6 for a given NER condition; the primary diagnostic for NER quality.

Algorithm 2 HTM Scoring — `compute_htm`

```

1 # Inputs: hypotheses H, segments S, graph G
2 # Output: macro-averaged HTM in [0, 1]
3
4 def htm_span(found, ref_label, ref_level, G):
5     if not found: return 0.0
6     node = G.get_by_name(found) or G.get_by_name(ref_label)
7     if not node: return 0.0
8     if node.level == ref_level: return 1.0 # exact tier
9     if G.same_branch(found, ref_label): return 0.5
10    return 0.0
11
12 scores = []
13 for row in H:
14     hyp = row.hyp
15     seg = S[row.id]
16     for term in unique_ner_terms(seg):
17         concept = G.ground(term, context=seg.fr)
18         if concept is None: continue
19         renderings = [concept.name] + G.candidates(concept)
20         found = first_phrase_in(hyp, renderings)
21         scores.append(htm_span(found, concept.name,
22                                concept.level, G))
23 return mean(scores) if scores else 0.0

```

Algorithm 3 HRA —
 compute_htm_hyp_vs_ref

```

1 # Inputs: hypotheses H, segments S, graph G
2 # Output: mean agreement in [0, 1] (HRA)
3
4 agreements = []
5 for row in H:
6     hyp, ref = row.hyp, seg.en_ref
7     seg = S[row.id]
8     for term in unique_ner_terms(seg):
9         concept = G.ground(term, context=seg.fr)
10        if concept is None: continue
11        renderings = [concept.name] + G.candidates(concept)
12        found_hyp = first_phrase_in(hyp, renderings)
13        found_ref = first_phrase_in(ref, renderings)
14        s_hyp = htm_span(found_hyp, concept.name,
15                          concept.level, G)
16        s_ref = htm_span(found_ref, concept.name,
17                          concept.level, G)
18        agreements.append(1.0 - abs(s_hyp - s_ref))
19 return mean(agreements) if agreements else nan

```

Algorithm 5 Hard Phrase Locking —
 force_words_ids (NLLB S5/S6)

```

1 # Inputs: segment seg, lock map L, graph G
2 # Output: constrained NLLB translation
3
4 phrases = []
5 for term in seg.terms:
6     concept = G.ground(term, context=seg.fr)
7     if concept is None: continue
8     en = L.get(term) or concept.name
9     phrases.append(en)
10
11 phrase_ids = [tok(p, add_special_tokens=False).input_ids
12               for p in phrases if p]
13 beams = min(10, 5 + len(phrases))
14 try:
15     out = model.generate(
16         **encode(seg.fr),
17         logits_processor=[PhraseLogitBoost(tok, phrases)],
18         force_words_ids=phrase_ids,
19         num_beams=beams, max_new_tokens=256)
20 except BeamSearchFailure:
21     # fallback: soft boost only (no hard lock)
22
23 out = model.generate(...,
24                       logits_processor=[boost])
25 return decode(out)

```

Algorithm 4 Soft Logit Boosting —
 PhraseLogitBoost

```

1 # Inputs: tokenizer tok, locked phrases P, boost delta
2 # Applied each decoding step to logits vector scores
3
4 class PhraseLogitBoost(LogitsProcessor):
5     def __init__(self, tok, phrases, boost=1.75):
6         self.boost = boost
7         self.ids = set()
8         for p in phrases:
9             for tid in tok(p, add_special_tokens=False).input_ids:
10                self.ids.add(tid)
11
12     def __call__(self, input_ids, scores):
13         for tid in self.ids:
14             if 0 <= tid < scores.shape[-1]:
15                 scores[:, tid] += self.boost
16         return scores
17 # S5-NLLB: delta=1.75; S5-Mistral: delta=1.25

```

Fluency: corpus BLEU [Papineni et al., 2002] and chrF++ [Popović, 2017] via SacreBLEU; neural: COMET [Rei et al., 2020]. HTM-A extends HTM matching across three tiers: T1 exact LLT; T2 adds Preferred Term (PT) parent; T3 adds all same-branch ancestors.

CCR and drift.. CCR = grounded/extracted spans; identical across S1–S6 for a given NER condition. Term drift = $(D - 1)/(N - 1)$ for D distinct renderings across N segments; drift = 0 is perfect consistency.

6 Results

6.1 Main Results

Under fine-tuned NER, graph-augmented Mistral systems close the fluency gap substantially (Table 4), confirming NER quality as the binding constraint.

HTM is recall-dependent on NER quality..

BLEU and chrF scores for S1 and S2 are identical across both NER conditions (Tables 3 and 4), confirming that the underlying translations did not change. However, HTM scores shift substantially: S1 HTM rises from 0.321 to 0.550 and S2 from

System	BLEU	chrF++	COMET	HTM	HRA
S1 NLLB	22.89	37.14	0.639	0.321	0.672
S2 Mistral	22.51	39.95	0.637	0.413	0.743
S3 GraphRAG	10.39	34.36	0.586	0.397	0.601
S4 rerank	9.93	34.24	0.583	0.429	0.579
S5 NLLB+boost	21.59	35.91	0.638	0.353	0.661
S5 Mist+boost	10.32	34.32	0.585	0.380	0.607
S6 NLLB+gls*	20.00	35.11	0.634	0.321	0.694
Reference	–	–	–	0.151	1.000

Table 3: Baseline NER condition (ner_biollm, CCR = 39.0%; 126 segments, 48_028 excluded). BLEU/chrF++ via SacreBLEU; COMET via wmt22-comet-da; HTM/HRA from Neo4j. *S6 is a glossary oracle (terms from reference).

System	BLEU	chrF++	COMET	HTM	HRA
S1 NLLB	22.89	37.14	0.639	0.550	0.429
S2 Mistral	22.51	39.95	0.637	0.735	0.476
S3 GraphRAG [†]	19.58	35.33	0.626	0.664	0.440
S4 rerank [†]	19.66	35.36	0.626	0.664	0.450
S5 NLLB+boost	21.37	36.06	0.637	0.578	0.403
S5 Mist+boost [†]	20.61	35.78	0.626	0.673	0.450
S6 NLLB+gls*	20.09	34.81	0.632	0.536	0.408
Reference	–	–	–	0.178	1.000

Table 4: Fine-tuned NER condition (ner_biollm_ft; CCR = 36.4%; 126 segments, 48_028 excluded). [†]Contamination-filtered (18/127 rows scored as empty hyps). *S6 uses a glossary oracle from the reference.

0.413 to 0.735 under fine-tuned NER. This is a *metric artifact*, not a translation improvement. HTM is computed only over spans the NER model successfully extracts; a stronger NER model exposes a larger and more challenging set of terminology spans, changing both the denominator and the matched set simultaneously. Scores across NER conditions are therefore not directly comparable and should be interpreted within each condition only.

6.2 CCR is the Binding Constraint

Table 5 isolates the effect of NER quality by holding all other pipeline components fixed and comparing the same system across NER conditions.

System	Δ BLEU	Δ chrF++	95% CI Δ BLEU
S1 NLLB	± 0.00	± 0.00	[0.00, 0.00]
S2 Mistral [‡]	± 0.00	± 0.00	[0.00, 0.00]
S3 GraphRAG	+7.07	+0.97	[+1.07, +11.10]
S4 rerank	+7.42	+1.12	[+1.18, +11.53]
S5 NLLB+boost	−0.16	+0.15	[−0.98, +0.66]
S5 Mist+boost	+7.16	+1.46	[+1.10, +11.66]
S6 NLLB+gls*	+0.09	−0.30	–

Table 5: Paired gain (fine-tuned minus baseline NER) on 126 segments. Δ BLEU is the paired bootstrap estimate (2,000 resamples, seed 42); 95% CIs for S3–S5-Mistral exclude 0. Δ chrF++ is the corpus-level paired difference. [‡]S2 shares the same inference JSONL across conditions. *S6 CI not computed (oracle glossary).

S1 is unchanged (no graph). S2 is exactly zero: the same inference JSONL is shared across both NER conditions, so Δ BLEU is zero by construction. S3–S5-Mistral show ≈ 7 -point Δ BLEU with intervals strictly above zero, isolating the NER bottleneck on the graph path. S5-NLLB is effectively flat. The S6 oracle glossary yields negligible gain, showing that logit boosting with reference-extracted terms does not improve over the graph path once NER quality is matched.

The 95% paired bootstrap confidence intervals for the BLEU gain of fine-tuned graph-augmented Mistral systems (S3–S5-Mistral) over their baseline-NER counterparts exclude zero (see Table 5), confirming statistical significance at $p < 0.05$. Graph-free systems (S1, S2) and S5-NLLB show confidence intervals that include zero, indicating their apparent gains are not reliably distinguished from noise.

S1 leads on corpus BLEU and S2 leads on chrF and HRA in the baseline NER condition (Table 3): graph-augmented systems fall well below both because the graph covers fewer than 40% of source spans while injected MedDRA strings hurt fluency in segments where it is active. S2’s lead on HRA (0.743) reflects the human-like output of full-document Mistral context without graph interference. The contrast between zero Δ BLEU for S1, S2, and S5-NLLB versus large Δ BLEU for S3–S5-Mistral isolates the bottleneck upstream of graph retrieval. Higher NER coverage would likely close the S2/S3 fluency gap.

A post-hoc CCR diagnostic (tools/ccr_diagnostic.py) classifying the 171 unique baseline NER terms against the Neo4j graph found that 60% are absent from

MedDRA entirely (NOT_IN_GRAPH), compared to 54% of the 246 fine-tuned terms; a further 6–8% are ambiguous (multiple concept matches, triggering the planner’s abstain policy), and only 23–32% ground via exact string match. The dominant not-in-graph category is populated by non-medical tokens (*pembrolizumab*, grade annotations, numeric ranges) that NER extracted as candidate spans but that have no MedDRA equivalent – confirming that the CCR ceiling is set by NER precision, not by graph coverage gaps.

6.3 Missing Terms, Flattening, and Consistency

Under baseline NER, S5-Mistral achieves the lowest missing-term rate; S1-NLLB misses the majority of grounded terms. Zero concept flattening events are observed for grounded L5 source terms, partly an evaluation artefact that HTM-A addresses. Cross-document modal agreement is highest for graph-free systems under baseline NER (S1: 0.993, S2: 0.984, S3: 0.946); under fine-tuned NER the gap narrows (S3: 0.976) but S2 (0.985) remains the most consistent Mistral system, confirming that the consistency advantage of S1-NLLB is a vocabulary artefact rather than terminological reasoning.

7 The HTM–Practice Gap

7.1 Why the Reference Scores Near Zero

On the baseline NER manifest, $r\text{HTM} = 0.151$ on the professional reference (Table 3); under fine-tuned NER the reference HTM is 0.178 (Table 4) because the grounded span set differs. In both cases the score stays below several MT systems on HTM. Professional translators do not transcribe MedDRA LLT strings verbatim: the LLT *Immune-mediated pneumonitis* appears as *pneumonitis* in tabular contexts and as *immune-related pneumonitis* in descriptive contexts.

7.2 Alias Expansion Does Not Close the Gap

Tier	Accepts	Low	Hi
T1	Exact LLT	0.151	0.178
T2	T1 + PT parent	0.151	0.178
T3	T2 + branch ancestors	0.151	0.178

Table 6: $r\text{HTM}$ -A on the professional reference under three alias tiers (Low = baseline NER manifest; Hi = fine-tuned NER manifest). All tiers yield identical scores within each manifest.

The ancestor chain for *pneumopathie inflammatoire* contains *pneumonitis*, so the null T1→T3 delta is not a graph coverage gap. Professional translators use phrasings such as *immune-related pneumonitis* and *inflammatory lung disease* that lie entirely outside the MedDRA preferred-label tree at any level. The mismatch is structural: MedDRA labels are designed for pharmacovigilance coding, not regulatory prose.

HTM should therefore be interpreted as a measure of MedDRA-proximity, not translation quality. The HRA column is the complementary measure. Notably, graph-augmented systems score consistently *lower* on HRA than S2 (Table 3: S3 = 0.601 vs. S2 = 0.743; Table 4: S3 = 0.440 vs. S2 = 0.476): the graph pushes outputs toward MedDRA labels that professional translators avoid. The fluency–terminology trade-off and per-system inference times are available as figures in the project repository ([results/ner_biollm/figures/](#)).

8 Consistency Failure Analysis

Term drift (qualitative).. *Pneumopathie inflammatoire* occurs in four baseline segments where it is grounded for S3/S4/S5-Mistral, and in eight fine-tuned segments (more NER hits for the same French string). Those systems emit two renderings—often *pneumonitis* in incidence-style sentences and *immune-mediated pneumonitis* elsewhere—yielding drift scores of 1/3 (baseline) and 1/7 (fine-tuned). S1 collapses to one rendering for this span in both conditions (drift 0). S2 is stable in baseline (drift 0) but shows non-zero drift under fine-tuned NER once the span appears in additional contexts. S3/S4/S5-Mistral thus exhibit the clearest cross-segment inconsistency for this exemplar.

Mean modal agreement (§6) shows the same pattern at corpus level: graph-free systems rank highest under baseline NER; graph-augmented systems recover partially under fine-tuned NER but do not overtake S2.

Translator language outside MedDRA.. Professional translators systematically prefer descriptive phrasings outside the MedDRA preferred-term vocabulary, falling into three patterns: **descriptive expansion** (e.g. “*immune-related pneumonitis*” for the PT PNEUMONITIS, scoring 0.0 under HTM at any alias tier); **hypernym substitution** (deliberate ascent to HLT/SOC for readability, penalised as a specificity error by HTM); and **compound**

paraphrase (clinical shorthand such as “*hand-foot syndrome*” lying outside the MedDRA branch entirely). Systems scoring well on HTM therefore use formal ontology labels that certified translators avoid, explaining the HTM–BLEU divergence. The appropriate evaluation lens is **HRA**, which measures agreement with professional translation practice rather than raw ontology proximity.

Unified expert evaluation.. To ground the system comparison empirically, we ran a unified expert evaluation across the same 30 segments as the register audit, covering all eight system variants (S1, S2, S3-base, S3-FT, S5-base, S5-FT, S6, human reference) using a GPT-4o senior EMA medical translator persona (seed 42; full results in `error_analysis/unified_eval.csv`).

The human reference dominated with 20/30 best-output votes (67%, average rank 1.50). Among MT systems, S2 (Mistral, full document context, no graph) achieved the highest average rank (3.27) and tied S3-FT for best-output votes (2/30 = 7% each), confirming that fluent contextual prose without ontology enforcement remains the closest MT approximation to certified translation practice. S5 received zero best-output votes in both NER conditions – consistent with the register mismatch finding that logit-boosted MedDRA label enforcement diverges from the clinical prose register EMA translators produce. The NER condition split was balanced (fine-tuned preferred in 10/30 segments, baseline in 10/30, tied in 10/30), indicating that NER quality determines performance within graph systems but does not override system architecture when all variants compete together. The evaluator flagged S1’s hallucination of *daxitinib* for *axitinib* as a critical pharmacovigilance error, and S3-base’s substitution of *Hypercarotinaemia* for *hypothyroidism* as a recurring contamination artefact from the baseline NER condition – both confirming that NER quality is a patient-safety concern, not merely a metric artefact. A pilot annotation (30 segments, 4 baseline systems) and a pilot expert evaluation (25 segments, 8 systems, different sample) motivated this design; the unified evaluation on the canonical 30-segment sample is the definitive result.

Mechanism.. *Pneumopathie inflammatoire* maps to two distinct MedDRA concepts (one of 7,679 ambiguous normalised keys in the graph). The planner commits no lock for ambiguous

terms, so S3/S4/S5-Mistral resolve the term context-locally per segment, producing drift scores of 1/3 (baseline) and 1/7 (fine-tuned). S1-NLLB produces zero drift not through terminological reasoning but through vocabulary rigidity. The fix is *soft ambiguity resolution*: commit the top-scoring candidate as a provisional lock for human review, consistent with CAT-tool workflows.

Output contamination.. In the fine-tuned NER condition, 18/127 segments per system (14.2%) contain prompt contamination: Mistral reproduced MedDRA metadata strings and instruction markers from its own system prompt. These rows were replaced with a null hypothesis for metric computation. Contamination biases raw scores in opposite directions—deflating BLEU (precision destroyed by injected strings) while inflating chrF (character overlap increased)—confirmed by S3 raw vs. filtered: BLEU 15.36→19.58, chrF 37.20→35.33. S1 and S2 are unaffected as they receive no graph context in their prompts.

9 Discussion and Limitations

The central finding is a paradox. S5 is more technically correct than the human reference by ontology standards – scoring higher on HTM – yet received zero best-output votes across 30 segments in the unified expert evaluation. The human reference dominated at 67%; **S2 (Mistral, full document context, no graph) is the best MT system** with an average expert rank of 3.27, ahead of S3-FT (4.69). S3-FT was competitive in an earlier pilot on a different 25-segment sample but does not replicate that advantage on the canonical sample. MedDRA labels are designed for pharmacovigilance database coding, not clinical prose; a translator writing “*immune-related pneumonitis*” is writing for a reader, while S5 writing PNEUMONITIS is writing for a database. Both are technically correct but serve different purposes, and optimising for HTM targets the wrong register.

The graph works, but only upstream of a CCR threshold. At 39% CCR the graph hurts fluency more than it helps terminology. Under fine-tuned NER (211 grounded spans vs. 184 baseline), graph-augmented Mistral systems gain ≈ 7 BLEU with bootstrap intervals excluding zero. The bottleneck was never graph retrieval or decoding strategy – it was always NER coverage. The S6 oracle glossary confirms this: even with reference-extracted terms, BLEU improves by only +0.09 points over S5-

NLLB.

Limitations.. *Single document and reference:* all results derive from one SmPC section against one professional translation. *HTM ceiling:* rHTM is low (0.15–0.18) and invariant to alias expansion, so HTM is only interpretable relatively. *NER artefact:* baseline NER contains hallucinated spans (*hypothyroïdie* in 92/127 segments). *Contamination:* 14.2% of fine-tuned Mistral graph outputs contain prompt contamination; filtered scores are used throughout. *Boost surface rate:* the lock map was not logged during inference; proxy analysis shows S5 surfaces NER terms verbatim in 34% of cases overall, dropping to 0% for multi-token MedDRA labels, confirming that term length is the hard ceiling on any enforcement approach.

10 Conclusion

TermPlanMT demonstrates that CCR is the binding constraint on graph-augmented regulatory MT: at 39% CCR, graph-free systems outperform all graph-augmented systems on fluency – S1 leading on BLEU, S2 on chrF and HRA – while improving NER quality yields ≈ 7 -point corpus BLEU gains for Mistral graph systems with bootstrap intervals excluding zero. A unified expert evaluation (30 segments, 8 systems, GPT-4o EMA translator persona) confirms that **S2 is the best MT system** by average expert rank (3.27), with the human reference dominating at 67% best-output votes. S5 received zero best-output votes in both NER conditions: the most ontology-correct system is never the preferred output, because MedDRA labels serve pharmacovigilance coding rather than clinical prose. Three broader findings emerge: MedDRA nomenclature and professional translation practice are structurally misaligned in ways alias expansion cannot fix; ambiguous grounding produces predictable term drift through the planning stage’s abstain-on-ambiguity policy; and NER quality is a patient-safety concern – S1’s hallucination of *daxitinib* for *axitinib* illustrates the risk of deploying low-quality NER in pharmacovigilance contexts. Substantially higher CCR with cleaner NER span boundaries is the necessary condition for graph-augmented systems to surpass S2; quantifying that threshold is left to future work with broader French biomedical NER.

Future work.. Graph-aware fine-tuning is the compelling next step: rather than prompting and

inference-time logit boosting, fine-tuning the LLM directly on serialised Neo4j subgraphs may teach it to embed MedDRA concepts naturally in clinical prose, bridging the register mismatch gap without hard token forcing. Substantially higher CCR (≈ 60 – 70%) with cleaner NER boundaries is a likely necessary condition for graph-augmented systems to surpass S2; quantifying that threshold with broader French biomedical NER is the direct next empirical step.

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MedDRA licence held under non-commercial academic agreement via MSSO/UPV/EHU, supervised by Gorka Labaka.

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Supplementary Figures

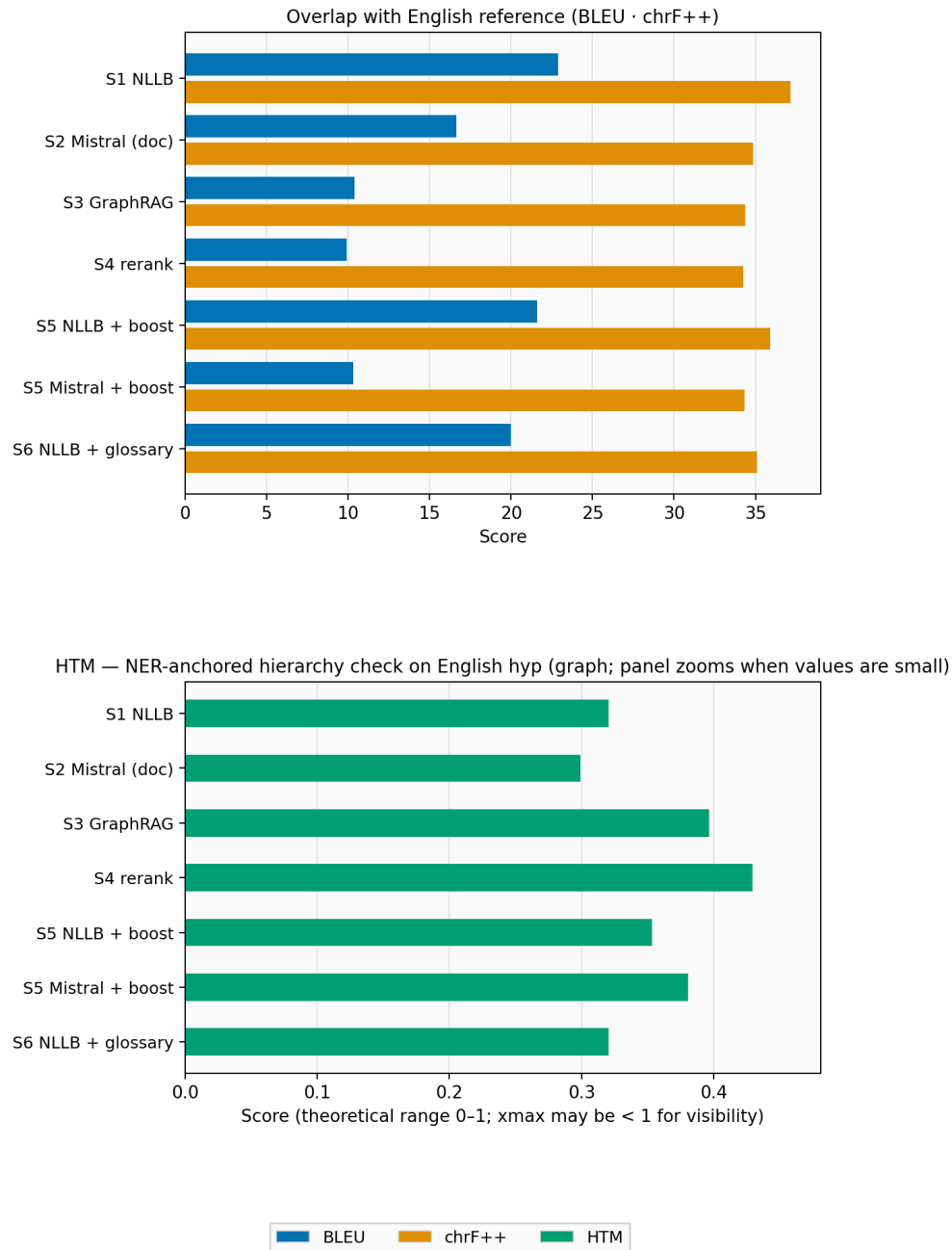


Figure 2: Full system comparison under baseline NER (ner_biollm, CCR = 39.0%). Top panel: BLEU and chrF++; bottom panel: HTM. S1/S2 lead on fluency; graph-augmented systems lead on HTM at the cost of BLEU, confirming the register mismatch.

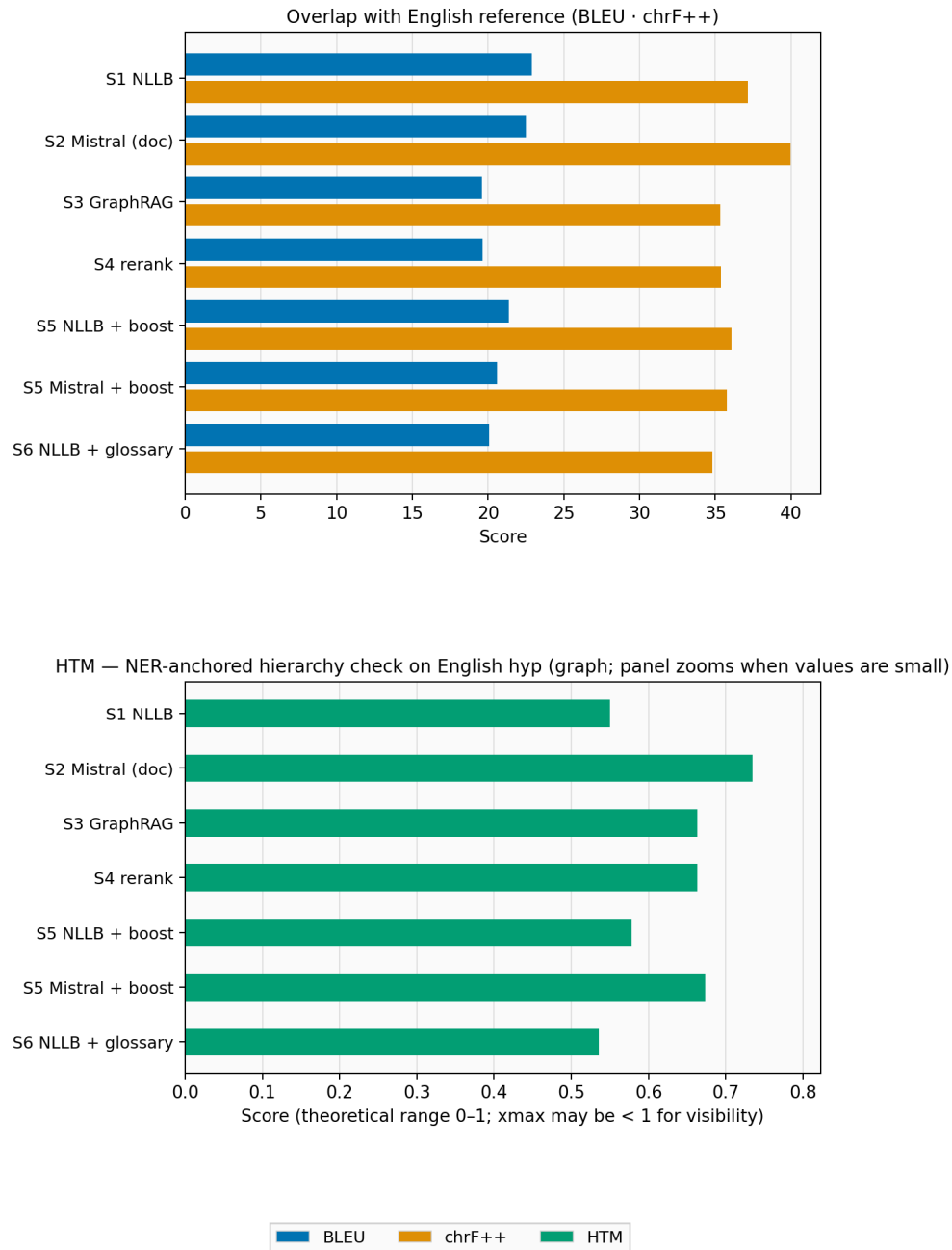


Figure 3: Full system comparison under fine-tuned NER (ner_biol1m_ft, CCR = 36.4%). Graph-augmented Mistral systems (S3–S5-Mistral) close the fluency gap substantially, confirming NER quality as the binding constraint.